

**AKHILA**

**Sr Data Analyst**

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**PROFESSIONAL SUMMARY:**

Overall 8+ years of experience as a data analyst with proficiency in gathering business requirements, Gap analysis, data quality assessment, data visualization and ETL testing in the healthcare domain.

- Worked on Cerner applications such as Citrix workspace, Relay and Health quest to access the patient data and billing information under CMS (1500) and UB04 format.
- Involved in evaluating the data according to regulatory standards like HIPAA, HITECH, HEDIS and SNOMED-CT.
- Proficient in Epic's suite of applications including EpicCare Ambulatory and EpicCare inpatient services.
- Demonstrated expertise in data privacy, data protection regulations, such as GDPR, CCPA and assisted in ensuring compliance through data governance practices.
- Proficient in defining data governance policies, standards, and procedures to govern the collection, storage, usage, and sharing of data assets.
- Proficient in FHIR to promote the interoperability among different modern health systems for seamless exchange of patient information.
- Proficient in claims processing, ICD-9 and ICD-10 coding systems, with expertise in analyzing and interpreting complex medical coding data.
- Demonstrated expertise in data profiling, data quality assessment, and data cleansing techniques to identify and remediate data issues and improve overall data quality.
- Extensive experience in gathering the functional and non-functional user requirements for creating the business requirements document (BRD) after participating in the JAD sessions.
- Supported cross-functional teams for the product developments using Agile methodology like sprint planning and user story mapping to manage product backlogs, prioritize features, and deliver iterative product updates on time and consistently meeting project milestones.
- Involved in creating the use cases and flow diagrams as per the requirements of the user.
- Experienced in Source to target data mapping to perform data migration using source files in formats CSV, JSON from SQL to GCP cloud using ETL pipelines.
- Experience in data visualization using Python, Tableau and R.
- Experienced working with informatica to extract, transform and load (ETL) the data for mapping of the data using source qualifier transformation.
- Skilled in data modeling and database design, with experience in creating and maintaining large, complex databases for healthcare organizations using STAR and snowflake schema.
- Experienced in building data pipelines using AWS glue and GCP data flow to automate the ETL process from legacy systems to cloud platforms.
- Designed and developed custom macros using VBA (Visual Basic for Applications) within the healthcare system to automate data entry, data validation, and report generation processes.
- Strong programming skills in Python, able to develop and automate data analysis workflows, build predictive models, and create visualizations using popular libraries like pandas, NumPy, and matplotlib.
- Proficient in creating executive-level dashboards and reports to communicate insights to healthcare providers, payers, and regulatory agencies, utilizing visualization tools such as Tableau and Power BI.
- Created Tableau scorecards and dashboards using stack bars, bar graphs, scattered plots, geographical maps to provide key information KPIs for decision making.
- Utilized Excel to remove duplicates, filter, and sort data, convert data types, split, or combine columns, and perform various data transformations to prepare the data for analysis.
- Experience in creating pivot tables, formatting, VLOOKUPS, and pivot charts to summarize and analyze large datasets, generate summary statistics, and create visual representations of data trends and patterns.
- Expertise in SQL including the ability to write complex queries SQL table joins, DDL, DML, PIVOTS views, constraints, and functions to SQL server.
- Created interactive dashboards in Tableau to visualize and analyze complex data sets, providing insights into key business metrics.
- Created numerous simple to complex SQL queries involving self-joins, correlated subqueries, stored procedures, functions, and triggers for diverse business requirements. Tuned and optimized queries by altering database structures and analyzing different query options to perform complex retrievals and analysis reporting tasks.

## TECHNICAL SKILLS:

|                              |                                                                                                |
|------------------------------|------------------------------------------------------------------------------------------------|
| <b>Analysis tools</b>        | My SQL, Microsoft SQL server, Tableau Desktop, Microsoft Azure, STATA, WEKA, Power BI Desktop. |
| <b>Data Analysis</b>         | Tableau (Desktop/Server/Reader), Python, R, MS-Excel.                                          |
| <b>Computer skills</b>       | Data cleaning, Data Mining, Data Visualization, Microsoft office                               |
| <b>Data modeling</b>         | Star and snowflake                                                                             |
| <b>Languages</b>             | bigdata, Datawarehouse environment, Python, R, SQL, SAS, XML                                   |
| <b>Cloud services skills</b> | AWS, GCP, Azure data factory and data bricks                                                   |
| <b>Tools</b>                 | Git, Jira, Informatica, STATA, WEKA                                                            |
| <b>Databases</b>             | ORACLE, MS Access, MY SQL, SQL server                                                          |
| <b>Operating systems</b>     | Windows, Linux, Mac OS                                                                         |
| <b>File formats</b>          | CSV, JSON, XML, AVRO, ORC, PARQUET                                                             |

## PROFESSIONAL EXPERIENCE:

**Client: CVS Health, Woonsocket, United States | Jun 2024 - Present**

**Role: Data Analyst**

### Responsibilities:

- Involved in the full HIPAA compliance lifecycle and conducted gap analysis to identify discrepancies between current system capabilities and desired functionalities and proposing solutions.
- Involved working closely with IT teams to analyze the Epic EHR and EMR system's issues related to data integration and data migration.
- Responsible for gathering relevant healthcare data from various sources such as electronic health records (EHRs), claims databases, and other healthcare systems.
- Developed and implemented data analysis strategies for healthcare cost analysis using HL7, FHIR, Medicare, and Medicaid datasets.
- Ensured compliance with HIPAA regulations and implemented necessary security measures to protect sensitive healthcare data.
- Analyzing and interpreting complex data sets in formats JSON and XML related to Medicare, Medicaid, and other healthcare claims to identify trends, patterns, and anomalies.
- Handled large sets of data in both structured formats and unstructured formats to perform the cleaning of the data.
- Expertise in performing data analysis, data validation, and quality assurance processes on FHIR datasets, ensuring accuracy and reliability.
- Data visualizations of scatter plot, countplot, heatmap using matplotlib, seaborn, plotly packages in python, outlier analysis using box plot
- Experienced using LOINC (Logical Observation Identifiers Names and Codes) to standardize and categorize healthcare data for analysis purposes.
- Collaborated with cross-functional teams and created the BRD by gathering functional and non-functional requirements and translating business needs into data analysis projects.
- Provided data profiling expertise and guidance to business stakeholders, supporting data-driven decision-making processes, and ensuring EDI data quality across the organization.
- Expertise in defining and establishing data governance policies, standards, and processes using Collibra, ensuring data quality, consistency, and compliance across the claims data ecosystem.
- Extensively worked with health quest and relay to access and analyze patient claims data for health care data analysis and developed validation techniques to assess the reliability of quality in patient data.
- Performed integration using HL7 and FHIR by integrating the EPIC data related to policy claims from disparate databases with other health insurance applications for underwriting and billing.
- Perform data collection, data cleaning, data wrangling, data analysis and building machine learning model on the data sets in both R and Python
- Implemented and managed EDI interfaces for data exchange with external partners such as payers, providers, and vendors.
- Developed Tableau workbooks to track and analyze sales performance, identifying trends and opportunities for revenue growth.

- Develop and maintain comprehensive quality reports, highlighting key performance metrics, SLA compliance, and potential risk areas to facilitate data-driven decision-making and enhance overall claims accuracy.
- Participated in software development life cycle methodologies like agile and played integral role in sprint planning sessions, contributing to backlog refinement, and tracking progress to ensure timely completion.
- JAD sessions are conducted involving the executive sponsor, stakeholders, and subject matter experts to identify and prioritize the requirements.
- Utilized Tableau Desktop to design and develop visually appealing reports and dashboards for stakeholders, improving data-driven decision-making processes.
- Created and maintained a BRD (Business requirement document) and gathered the functional and non-functional requirements based on business requirements of the user.
- Involved in creating use cases and flow diagrams and collaborated with development teams to ensure smooth integration of the system and perform user acceptance testing (UAT) to validate the implemented functionalities.
- HL7 messaging standard is used to integrate both EHR and EMR with other related healthcare information technology systems.
- Mapped and translated diagnosis and procedure codes between ICD-9 and ICD-10 to facilitate data analysis and reporting.
- Collaborated with cross-functional teams to gather requirements and translate business needs into Tableau visualizations and reports.
- Collaborated with business stakeholders and subject matter experts to understand data mapping requirements for source to target data mapping and ensure alignment with business needs.
- Utilizing JIRA project management tool to track project progress, document issues and provided regular status updates to project team.
- Possessed exceptional analytical skills and a deep understanding of SQL and ETL processes, enabling accurate data analysis and extraction of insights.
- Demonstrated proficiency in ETL tools like Talend and SSIS, leveraging them to efficiently transform and load complex data sets.
- Exhibited expertise in Snowflake SQL, enabling efficient querying and manipulation of data within the Snowflake cloud data platform.
- Utilized SQL expertise to efficiently retrieve relevant data for claims analysis, supporting efficient claims processing and resolution.
- Experienced using big query in GCP cloud to manage large sets of data and migrated the data into SQL server management studio.
- Snowflake schema was implemented for non-redundancy in data base.
- Experienced working with ETL tools like SSIS by using source qualifier transformation for data mapping.

**Environment:** Medicare, Medicaid, HIPAA, EHR, EMR, HL7, CMS, AWS, GCP, SSIS, Azure data bricks, Agile, Waterfall, S3, SAS, ETL, Informatica, snowflake schema, SQL server, power BI, tableau.

**CBRE, Texas, United States | Mar 2022 – May 2024**

**Role: Sr. Data Analyst**

**Responsibilities:**

- Collected and gathered vast amounts of patient data from diversified sources to evaluate the trends and recognize the areas for improvement.
- Collaborated with product owners, scrum masters, and development teams to prioritize data analysis tasks, plan sprints, and ensure timely delivery of insights to stakeholders as a part of agile SDLC.
- Conducted Joint application development (JAD) sessions for developing business system requirements and created the BRD (Business requirement document) by gathering the business needs and requirements of the user.
- Utilized ETL tools such as SQL Server Integration Services (SSIS) and Visual Studio to create and manage complex data integration workflows and transformations.
- Using SQL and oracle, extracted, manipulated, and analyzed the data stored in different databases to build the insights.
- Worked on Python while using Apache Spark (PySpark) while working on Big Data management including building Spark data frames, Spark SQL etc.

- Conducted extensive data exploration and analysis on large-scale healthcare datasets to identify relevant factors and patterns related to disease prediction.
- Demonstrated exceptional analytical skills coupled with a strong foundation in SQL and ETL processes, facilitating in-depth data analysis and insightful interpretation.
- Exhibited proficiency in ETL tools such as Talend and SSIS, leveraging them to efficiently extract, transform, and load complex healthcare data.
- Proficient in integrating with Microsoft Excel, enabling users to access and analyze data directly within Excel spreadsheets.
- Analyzed large sets of patient data by using data warehouses like snowflake and data bricks mounted on AWS, GCP and AZURE.
- Created various kinds of charts to visualize the data using tableau and analyzed the claims requests by using the advanced features by calculating fields, R integration, parameters, and table calculations.
- Created the interaction between the visuals by using several advanced features like filtering, sorting, and grouping the data in POWER BI.
- Developed the functional and non-functional requirements according to the business user requirements.
- Automated reporting processes using Tableau Server, reducing manual effort and improving efficiency.
- Conducted assessment to ensure data quality and completeness to acknowledge data quality issues with validation techniques.
- Participated in migration of data from source to target data mapping, data modeling during analysis of data.
- Experience in using excels formula language and functions allowing analysts to perform complex calculations and build models.
- Utilized expertise in Snowflake SQL to seamlessly query and manipulate data within the Snowflake cloud data platform, supporting accurate disease prediction analyses.
- Utilized SQL proficiency to efficiently extract relevant data for disease prediction modeling, supporting accurate and timely insights.
- Involved in managing the data integration and ensuring the data quality across AWS and GCP cloud services.
- Involved in creating the use case stories and conducted user acceptance testing (UAT) and documented the rules.
- Track and manage project progress using JIRA, ensuring that timelines and milestones are met.
- Measured the effectiveness of the program by advancing the Key performance indicators (KPI's) in POWER BI.
- Generated POWER BI and TABLEAU dashboards to analyze the outcomes of system and check system effect for each condition and compared the reports prior and after using the system for stakeholders to analyze and enhance the outcomes of the system.
- Demonstrated proficiency in writing complex SQL queries, utilizing JOINS, subqueries, and aggregate functions to extract, transform, and analyze data effectively.
- Proficient in crafting complex SQL reports and views for generating meaningful business insights and aiding in data-driven decision-making.
- Performed Decision Tree Analysis and Random forests for strategic planning and forecasting and manipulating and cleaning data using dplyr and tidyr packages in Python.
- Implemented Tableau best practices to optimize dashboard performance and ensure data accuracy and consistency.
- Utilized statistical methods such as regression analysis, time series analysis, and clustering to extract meaningful insights from diverse medical data sources.
- Designed and implemented Extract, Transform, Load (ETL) processes using Informatica to integrate disparate healthcare data sources into a centralized data warehouse.
- Involved in writing complex SQL queries using SQL server management studio (SSMS) to preprocess and clean the data to avoid duplicates and null values.
- Proficient in Hive, a data warehouse infrastructure built on top of Hadoop, for querying and analyzing structured and semi-structured data.
- Developed custom Python modules (pandas and NumPy) and functions to support specific data processing requirements, ensuring code reusability and maintainability.
- Leveraged T-SQL's analytical functions, subqueries, and common table expressions (CTEs) to perform advanced data manipulations and aggregations.
- Created custom SAS macros for repetitive tasks, resulting in increased efficiency and reduced processing time for disease prediction analysis.

- Utilized SAS SQL to perform complex queries and aggregations, enabling the extraction of valuable insights from the patient data.
- Processed the Raw data from CSV files in to organized form by applying Data Cleaning techniques using pandas and NumPy on Python.

**Environment:** AWS, GCP, Agile, Waterfall, SAS, Informatica, snowflake schema, JIRA, ETL, Python libraries, SQL server, power BI, tableau.

### **Value Labs, Hyderabad, India | | Sep 2019 – Dec 2021**

#### **Role: Data Analyst**

#### **Responsibilities:**

Utilized Agile SDLC to manage and prioritize multiple projects, ensuring timely delivery and high-quality results. Developed and maintained comprehensive source to target data mapping documents, including field-level mappings, transformation rules, and data dependencies.

Utilized APIs to automate repetitive tasks, streamline data workflows, and integrate different systems.

Formulated procedures for data extraction, transformation, and integration of health care data using ETL/ELT pipelines.

Analyze the price and rate trends of contingent labor data points using: Excel, Tableau.

Perform variance analysis comparing current market contingent labor costs versus Manpower client program labor costs.

Provide cost estimates on labor under different model scenarios to clients to help them achieve the right savings versus quality/efficiency.

Use Tableau to create effective data visualizations of rate trends over time and other data points

Present findings and consult with internal/external stakeholders including heads of IT, HR and procurement at major corporations.

Make pricing recommendations to clients along with job taxonomy recommendations.

Maintain and update data repositories.

Analyze large data sets, extracting key data points to tell a holistic story around rates and program metrics.

Manage one employee, training the employee, managing their workload and checking their work.

Currently managing a joint project with IT to create a software/web based rate platform that will provide global pay and bill rate data to clients and internal teams. This project will allow SQL reporting. The database is MS SQL Server.

Helped develop a new pricing model for field teams to extract labor pricing by job title and location thus saving my team and the field teams' time by giving them another source of data to use.

Developed complex transformation logic using ETL tools like Talend and SSIS to ensure seamless migration of diverse healthcare data formats.

Developed SQL scripts to automate data validation processes, ensuring efficient and consistent verification of migrated data.

Developed complex SQL queries to extract, transform and analyze healthcare data, ensuring data integrity and accuracy by identifying and resolving data inconsistencies, missing values, and outliers using T- SQL procedures and triggers.

Performed ad-hoc data analysis to support strategic decision-making processes, providing insights on cost-saving initiatives, reimbursement optimization, and budget forecasting.

Conducted thorough data validation and User Acceptance testing with QA team for identifying and resolving data discrepancies and anomalies.

Created visually appealing and interactive reports and dashboards using tools like Power BI and Tableau to present healthcare cost analysis findings to stakeholders.

**Environment:** Agile, Waterfall, ETL, Informatica, EDI Data Lake, Data warehouse, SQL, ad-hoc, Power bi, Tableau.

### **PrimefactorX, Hyderabad, India | | Jun 2016 – Aug 2019**

#### **Role: Data Analyst**

#### **Responsibilities:**

- Gathered, clean, and validated financial data from multiple sources, including market data providers, regulatory filings, and internal systems.
- Ensured compliance with regulatory requirements such as FCRA (Fair Credit Reporting Act), FINRA (Financial Industry Regulatory Authority), and SEC (Securities and Exchange Commission).
- Participated in the full lifecycle implementation of ERP systems, including requirements gathering, system design, testing, training, and post-implementation support.

- Conducted business process mapping and analysis to identify opportunities for process improvement and optimization within the ERP environment.
- Experience in waterfall software development life cycle (SDLC) methodology for delivering data analysis projects in a sequential manner.
- Created Business Requirements Documents (BRD) involving stakeholders for gathering the requirements to meet the aligned expectations.
- Involved in identifying the functional and non-functional requirements and test cases based on the business requirements.
- Involved in creating the facts and dimension tables for storing the data in SQL server database.
- Developed data dictionaries and designed star and snowflake schemas and used ETL for data integration and used project management tool like JIRA for creating the user stories.
- Develop and maintain detailed documentation of source to target data mappings, including data transformations and business rules.
- Performed SQL queries to clean the data for removing the duplicates and used SQL server for writing SQL queries to extract data from various sources into a centralized data warehouse.
- Experienced working with SSIS to streamline data processing and improve accuracy.
- Conducted User Acceptance Testing (UAT) with quality assurance (QA) team to ensure the system's functionality has met the user needs and quality standards.
- Developed interactive dashboards and reports using tools such as POWER BI and tableau to visualize clinical decision support system performance and trends.

**Environment:** FCRA, ERP, SEC, AWS, GCP, Azure, Agile, Waterfall, JIRA, Informatica, Talend, snowflake schema, JIRA, SQL server, power BI, tableau.